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Mechanical and Aerospace Engineering

Introduction and Motivation

- Long-Term Objective:** Support autonomy for individuals with disabilities by enabling independent feeding and reducing reliance on caregivers.
- Motivation:** There is no cohesive, documented, and publicly accessible meal assistance feature for the Stretch 3 assistive robot.
- Project Goal:** Develop an electromyography (EMG) based controller for the Stretch 3 to provide a user-friendly, open-source, safe, customizable, and intuitive assistive feeding system.

Methods

EMG Subsystem: Developed a real-time EMG processing system to command the Stretch robot to move. Tested digital filters and implemented threshold initialization sequence.

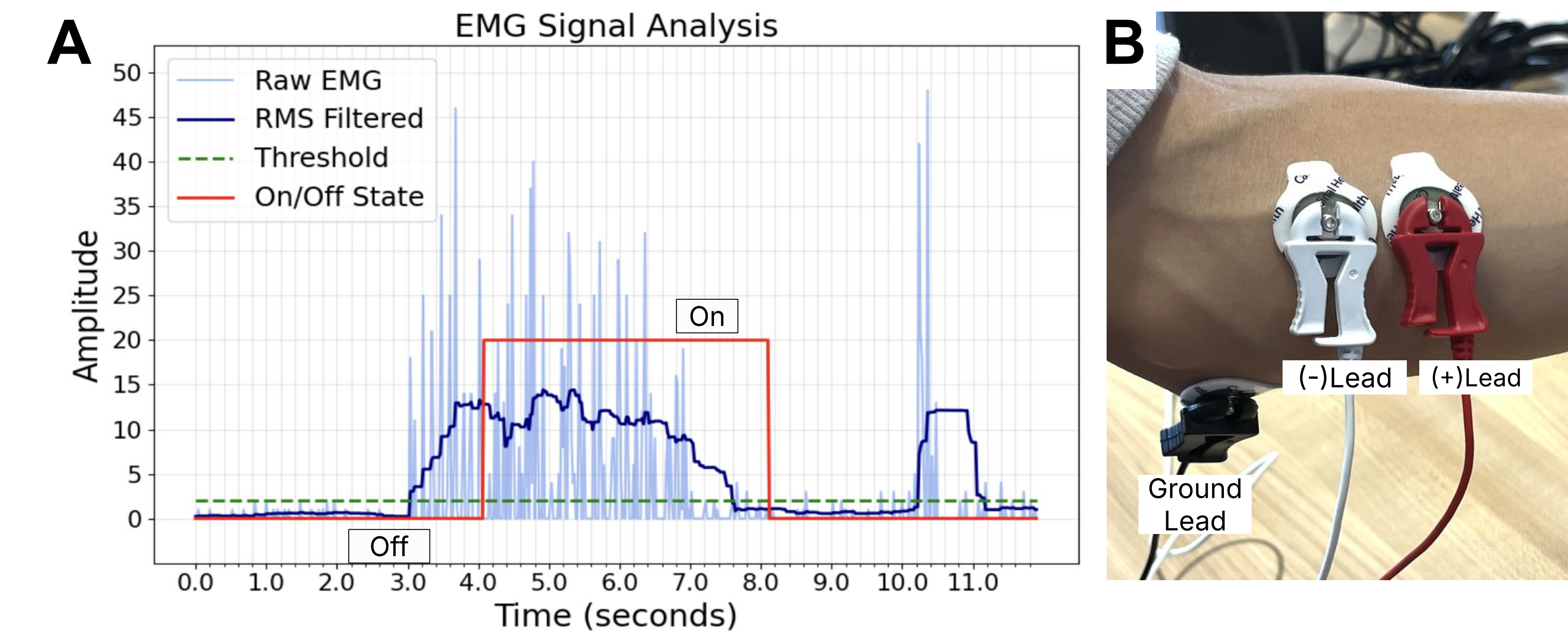


Figure 1. (A) Real-time flexor carpi radialis EMG data collected via Arduino and digitally filtered with Root Mean Square function **(B)** Selected EMG sensor placement for comfortable and intuitive activation.

Robot Subsystem: Developed custom code that integrates EMG signals with robot trajectory movements, prompting the Stretch 3 mobile manipulator robot to perform a custom feeding task.

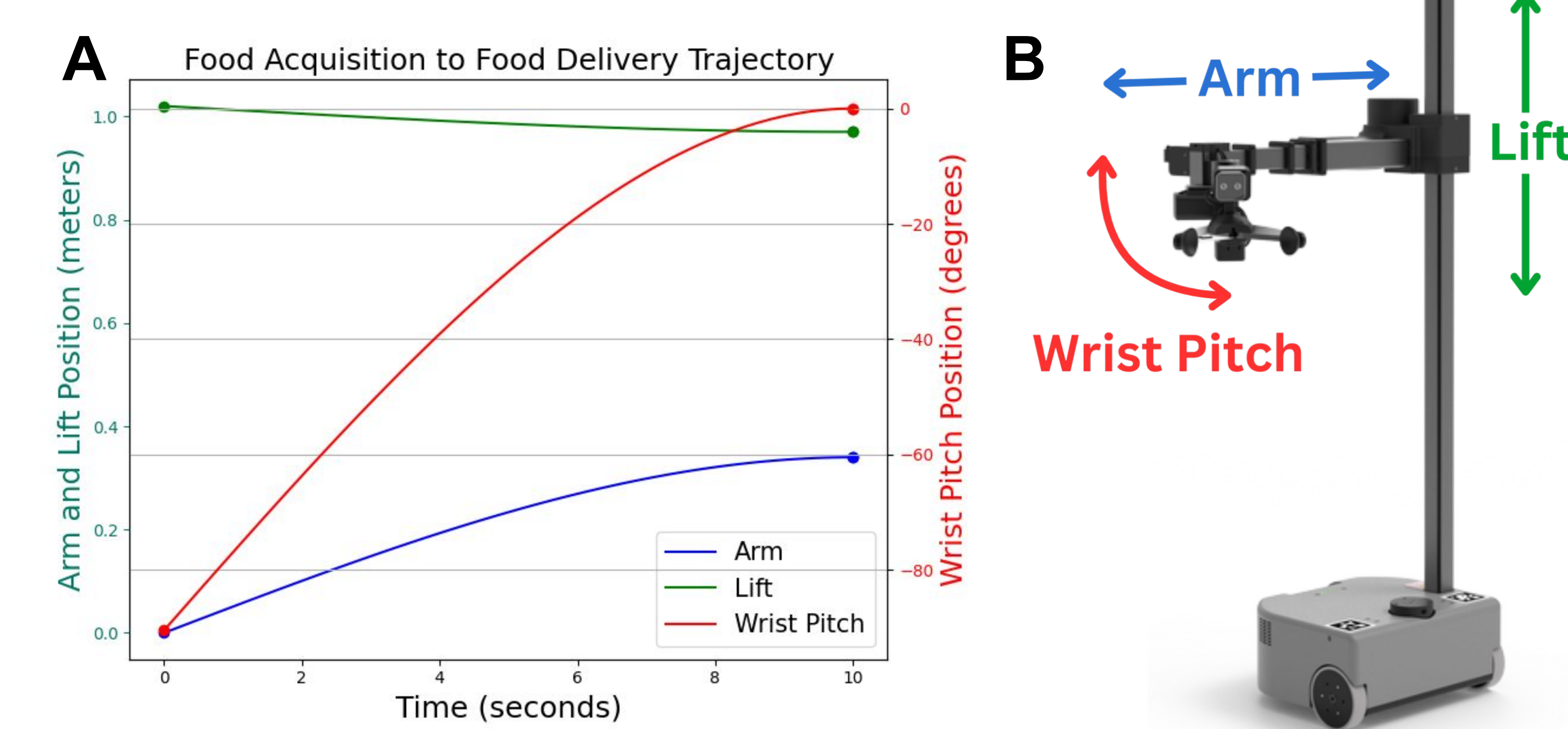


Figure 2. (A) One of three of the robot's spline trajectories of all synchronous joint movements during a feeding motion. **(B)** Diagram of the Stretch 3 robot with arrows indicating the corresponding joint movement directions.

Results and Discussion: Integrated Electromyography and Stretch 3 System

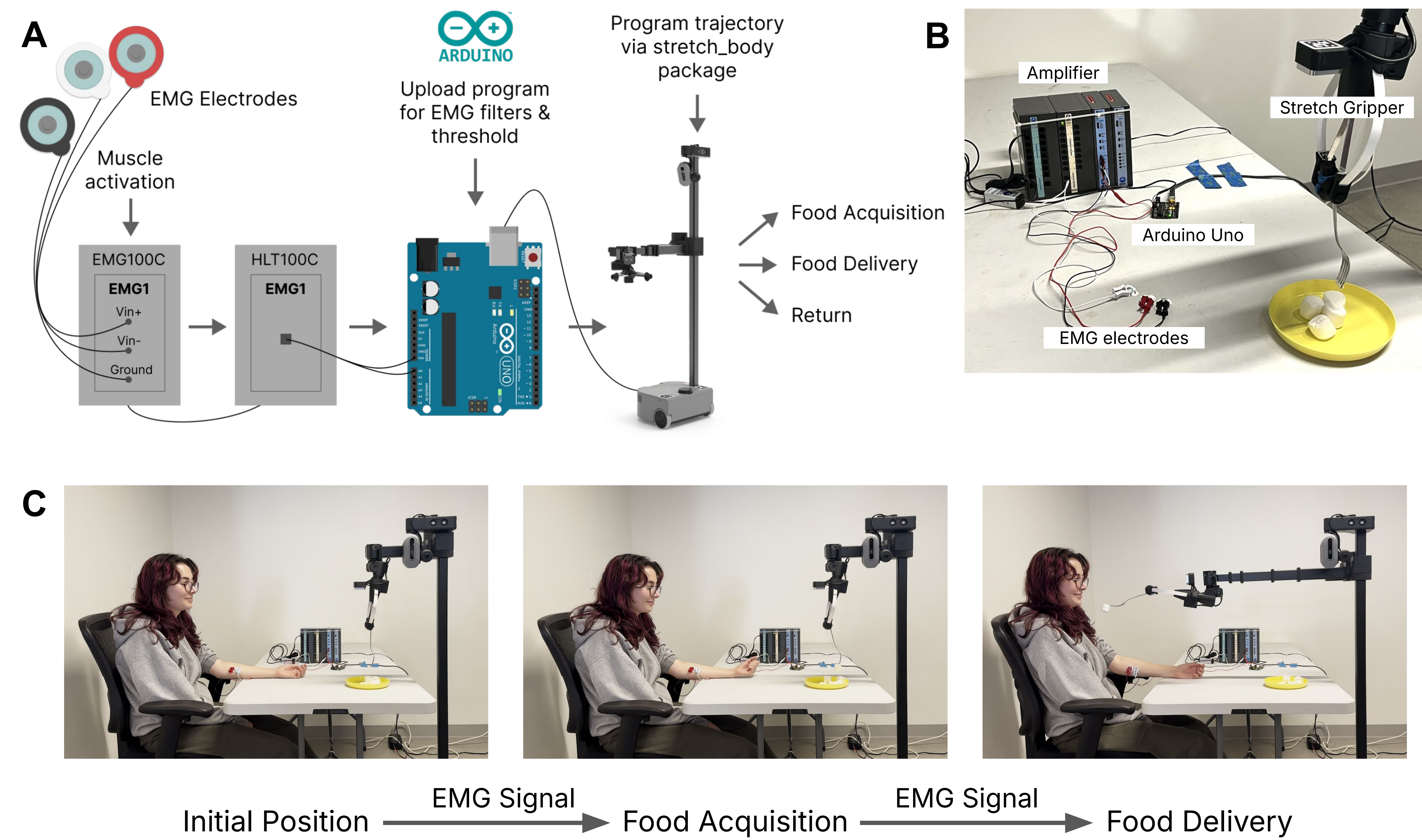


Figure 3. (A) Schematic for overall system workflow with EMG detection input, signal processing, and robot output. **(B)** Final setup for integrated EMG sensors and Stretch 3 Robot. **(C)** Three primary poses detailing the full robotic trajectory. Each activation signal from muscle contraction triggers predefined movement between successive poses before waiting for a subsequent command upon arrival.

Conclusions

- Developed proof-of-concept robotic system for assistive feeding with personalized EMG-controlled manipulation to acquire food and serve the operator, supporting varied levels and locations of muscular activation capability.
- Laid the groundwork for a Stretch robot control program to be used and developed by any user, to help bring greater independence for those with upper limb mobility impairments.

Future Directions

- Hardware:** Integrate second EMG input channel for advanced control, refine end effector to improve food acquisition, and add accessible emergency stop function.
- Testing:** Conduct trials with diverse participants to refine usability across different abilities. Integrate customizable GUI that adapts to each individual's feeding preferences.

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